

What is neurovascular coupling and uncoupling and how does it relate to the limitations of current neuroimaging methods, specifically EEG and fMRI. Given these limitations, what aspects of neuroimaging today should we aim to improve upon when developing new imaging techniques?

The State of Current Methods: EEG and fMRI

In lecture, we discussed the importance and most common research techniques in systems neuroscience, the study of how elements of our nervous system work together to generate a function. Systems neuroscience has developed various methods and techniques for recording and determining neural activity: functional neuroimaging techniques like fMRI, PET, fNIRS, and FUS provide indirect measures of neuronal activity while EEG and MEG provide more direct measures of electrical neural activity. Electrical activity is driven by neuronal changes in charge caused by neurons firing action potentials. Electroencephalograms (EEG) picks up these changes in electrical signals through electrodes placed on the scalp. While EEG is easy to perform on humans due to its non-invasive nature and is beneficial for its high temporal resolution, EEG's poor spatial resolution makes it difficult to know the source of the signals that the EEG is recording. Functional magnetic resonance imaging (fMRI), a non-invasive method of neural imaging which measures blood oxygenation levels in the brain, improves EEG's low spatial resolution at the cost of low temporal resolution. So, while we can visualize more specific areas where activity occurs, we lose the ability to see rapid or fleeting changes in brain activity.

The Basis of fMRI: Neurovascular Coupling

It's important to reiterate that fMRI indirectly measures brain activity. fMRI measures blood flow rather than neural activity and, "The efficiency of fMRI in studying neural processes in the human brain depends on the precise coupling between neural activity and blood flow".¹ Neural and metabolic activity in the brain have a close relationship that make fMRI a great method for neural imaging: during neural activity, oxygen and glucose are consumed and must be subsequently replenished to carry out cellular processes like neurotransmitter recycling or the restoration of an ionic gradient after an action potential. To replenish oxygen levels, the cerebral blood flow in the just previously active region increases.² Thus, an increase in neural activity also means a subsequent increase in cerebral blood flow. This neurovascular coupling allows techniques like fMRI to indirectly record neural activity through metabolic activity.

A Potential Blindspot: Neurovascular Uncoupling

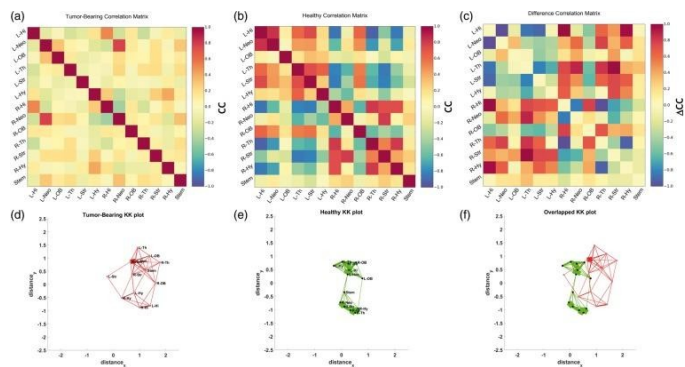
Still, fMRI's indirect nature of the measurement of neuronal electrical activity leaves us wondering if there are instances where neuronal activity does not inspire change in cerebral blood flow. In fact, researchers claim that "this coupling is not perfect".³ Researchers Pillai and Mikulis have suggested that in patients with certain neural abnormalities, lesions, impaired cortices, or cerebrovascular disease, neurovascular uncoupling (NVU) is possible. If NVU occurs, blood oxygen level dependent (BOLD) fMRI tasks may present false-negative activation levels despite there being positive levels of electrical neuronal activity.⁴ These false-negative activations levels in neurally impaired patients can be detrimental for clinical and presurgical fMRI scans. Essentially, one of the major benefits of fMRI, good spatial resolution, is diluted by an inability to record uncoupled neural activity. A study done by Pak et al. at the Division of Cancer Imaging Research at Johns Hopkins suggested that, "recent preclinical and clinical studies have demonstrated that brain tumors can exhibit NVU that can confound the interpretation of fMRI data".⁵ In tumored brains, for both resting state and task based fMRI recordings, NVU has potentially impaired or decreased the BOLD fMRI response. Figure 1 displays (a) the resting state functional connectivity of regions of interest (ROI) from a brain tumor-bearing mouse, (b) the resting state functional connectivity of ROI in a healthy mouse, and (c) the difference matrix of connectivity between the tumor bearing and normal mouse ROI. The differences in functional connectivity

between regions was lost in the tumor-bearing mouse as a result of NVU producing the highly varied difference between Figure 1a and 1b. Furthermore, evidence from Pak et al.'s study even suggest that brain tumors exert a more global effect, rather than one localized to the tumor area, on functional connectivity in the brain; in this case, a brain affected by cancer may produce fMRI images that indicate overall decrease in functional connectivity that can lead to cognitive deficits, but it seems unclear that this is an accurate representation of activity in the brain given the tumor-induced NVU.⁵ Referencing Figure 1 again, the BOLD signal fluctuations in many ROI in tumor-bearing mice were notably diminished (Figure 1b) compared to the BOLD signal fluctuations visualized in Figure 1a.

Figure 1. Brain tumors disrupt resting state functional connectivity levels and BOLD signal fluctuation due to NVU⁵

Implications of NVU on accuracy of current techniques

While researchers are currently debating the implications of NVU on interpreting BOLD signals and fMRI data, the question of whether or not our current imaging techniques are accurate and full accounts of our neuronal activity remains. The limitations of EEG's spatial resolution capacity and fMRI's temporal resolution capacity and, now, impacted spatial resolution as a result of NVU leaves us wondering if there's information about the brain we simply just aren't equipped to capture yet. Researchers at the Center for Functional MRI at UCSD agree that, "the BOLD response alone is ambiguous", but offer models for interpreting the BOLD signal in new ways and emphasize that fMRI methods provide very specific measurements of neural activity, specifically measurements of cerebral blood flow and oxygen metabolism. The variability of these responses in different scenarios should be studied further in order to better understand how they affect our understanding of neural activity from these measurements.⁶ I realize that suggesting there may be neural activity we don't yet have the technology to record may seem like an ill-posed and unprovable problem, but NVU gives us direction, justification, and motivation for improving our recording methods and techniques in systems neuroscience.



New alternatives & developments in neural imaging

Since 2009, researchers who understand the indirect link between the BOLD signal and neuronal activity have been acquiring EEG and fMRI data simultaneously to record neural activity.^{7,8} While we don't know if this combined technique solves the neural activity recording issues associated with NVU, the combination of these techniques allows researchers to study brain function with higher temporal and spatial resolution than just one of the techniques alone.⁷ Furthermore, a study by Nunez-Elizalde et al. investigated another imaging method called functional ultrasound imaging (fUSI). fUSI, although still reliant on neurovascular coupling, was found to have a higher spatial resolution than fMRI and, through simultaneous neural activity and fUSI signal recordings, to have strong correlations with local neuronal firing rates.⁹ Overall, there seems to be support for the idea that the limitations to our current methods and techniques for recording neural activity give impetus for the continued questioning of our current interpretations of BOLD fMRI signals, as well as research and development of new recording techniques.

WORKS CITED

1. Leopold D. A. (2009). Neuroscience: Pre-emptive blood flow. *Nature*, 457(7228), 387–388.
<https://doi.org/10.1038/457387a>
2. Brian N. Pasley and Ralph D. Freeman (2008) Neurovascular coupling. *Scholarpedia*, 3(3):5340.
3. Goyal, M. S., & Snyder, A. Z. (2021). Uncoupling in intrinsic brain activity. *Proceedings of the National Academy of Sciences of the United States of America*, 118(34), e2110556118.
<https://doi.org/10.1073/pnas.2110556118>
4. Pillai, J. J., & Mikulis, D. J. (2015). Cerebrovascular Reactivity Mapping: An Evolving Standard for Clinical Functional Imaging. *American Journal of Neuroradiology*, 36(1), 7–13.
<https://doi.org/10.3174/ajnr.a3941>
5. Pak, R. W., Hadjiabadi, D. H., Senarathna, J., Agarwal, S., Thakor, N. V., Pillai, J. J., & Pathak, A. P. (2017). Implications of neurovascular uncoupling in functional magnetic resonance imaging (fMRI) of brain tumors. *Journal of cerebral blood flow and metabolism : official journal of the International Society of Cerebral Blood Flow and Metabolism*, 37(11), 3475–3487.
<https://doi.org/10.1177/0271678X17707398>
6. Buxton, R. B., Valerie, Simon, A. B., Moradi, F., & Amir Shmuel. (2014). Variability of the coupling of blood flow and oxygen metabolism responses in the brain: a problem for interpreting BOLD studies but potentially a new window on the underlying neural activity. *Frontiers in Neuroscience*, 8.
<https://doi.org/10.3389/fnins.2014.00139>
7. Mulert C. (2013). Simultaneous EEG and fMRI: towards the characterization of structure and dynamics of brain networks. *Dialogues in clinical neuroscience*, 15(3), 381–386.
<https://doi.org/10.31887/DCNS.2013.15.3/cmulert>
8. Maria João Rosa, Kilner, J. M., Blankenburg, F., Josephs, O., & Penny, W. D. (2010). *Estimating the transfer function from neuronal activity to BOLD using simultaneous EEG-fMRI*. 49(2), 1496–1509.
<https://doi.org/10.1016/j.neuroimage.2009.09.011>
9. Nunez-Elizalde, A. O., Krumin, M., Reddy, C. B., Montaldo, G., Urban, A., Harris, K. D., & Carandini, M. (2022). Neural correlates of blood flow measured by ultrasound. *Neuron*, 110(10), 1631–1640.e4.
<https://doi.org/10.1016/j.neuron.2022.02.012>